

An Exact Computer-Assisted Lower Bound for Packing 17 Unit Squares in a Square

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<https://github.com/Mira-acc/17squares>

September 2026

Abstract

Let $s(17)$ be the least side length of a square containing seventeen pairwise interior-disjoint unit squares, with arbitrary orientations. We give an exact computer-assisted certificate proving $s(17) > 4.613028635886$. A nonnegative rational measure on 1,620 points has total mass $16.99798498 < 17$, yet assigns mass at least one to the interior of every contained unit square. An exact sweep verifies all translations of smaller closed cores at 2,881 directions; a strict containment inequality covers the intervening orientations. Rational scaling gives a slightly larger radical lower bound with a weak inequality. The two accumulation engines share one geometric kernel. The earlier sixteen-point proof of $s(17) > 4.468292$ is retained in an appendix.

1 Result and provenance

Write $Q_L = [0, L]^2$. Squares in a packing may touch, but their interiors must be disjoint. The minimum defining $s(17)$ is attained: centres and orientations range over a compact parameter space, and containment and nonoverlap are closed conditions.

Theorem 1. *Let*

$$L = \frac{4613}{1000}, \quad B = \frac{19997}{20000}, \quad h = \frac{207107}{1440000000}.$$

Then

$$s(17) \geq S_* := \frac{L\sqrt{1+h^2}}{B(1+h)} = \sqrt{\frac{17650291964463886688094912400}{829429719507765981945905041}} > 4.613028635886.$$

The base certificate excludes a packing at side 4.613. The endpoint $S_* = 4.613028635886110\dots$ follows by dilation; exclusion at that endpoint is not asserted. The cited upper-bound construction is Bidwell's packing, of side about 4.67553009360455[1, 3]. Thus the resulting interval is

$$4.613028635886 < s(17) \leq 4.67553009360455\dots$$

This is a claim about the supplied certificate and cited comparison data, not an exhaustive assertion of priority over other work.

The starting 1,184-atom certificate at side 4.59 is due to Joshua Levy's *squares* project[4]. The weighted covering principle and covering linear-program strategy predate this calculation, and the sharp dilation lemma used below is credited to Levy's T-022 note[5]. The preceding weighted certificate proved $s(17) > 4.607028598640$. The present continuation raises its base side from 4.607 to 4.613 by changing the spatial support and weights. The core side, direction net and exact geometric kernel are unchanged. The earlier certificate and its Levy/Mira support provenance remain in `certificates/lower_bound_4p607/`. The new certificate has 206 positive symmetry orbits, expanding to 1,620 atoms.

2 Weighted interior coverage

Let $\mu = \sum_j w_j \delta_{p_j}$, where $p_j \in Q_L$ and $w_j \geq 0$ are rational. If $\mu(Q_L) < 17$ and $\mu(\text{int } U) \geq 1$ for every contained unit square U , then seventeen interior-disjoint such squares would have total interior mass at least 17 and at most $\mu(Q_L)$, a contradiction. Sixteen unavoidable points, each of weight one, are the historical special case in the appendix.

The new certificate is invariant under the eight symmetries D_4 of Q_L . This restricts the measure, not a hypothetical packing. Set

$$T = \frac{207107}{500000}, \quad m = 2880, \quad t_k = \frac{kT}{m}, \quad \theta_k = 2 \arctan t_k \quad (0 \leq k \leq m).$$

The checker verifies nonnegative weights, atom containment, D_4 invariance, total mass below 17, and

$$T^2 + 2T \geq 1, \quad B^2(1+h)^2 < 1+h^2, \quad h = T/m < 1. \quad (1)$$

It then checks that every closed side- B square contained in Q_L at each θ_k has mass at least one.

Container symmetries and square symmetries reduce any individual orientation to $[0, \pi/4]$. The first inequality in (1) ensures that the net reaches this interval's upper endpoint. For the nearest net angle, the angular distance δ satisfies

$$\tan \delta \leq \frac{t_{k+1} - t_k}{1 + t_k t_{k+1}} \leq h.$$

For $0 \leq z \leq h < 1$ the identity

$$(1+h)^2(1+z^2) - (1+z)^2(1+h^2) = 2(h-z)(1-hz) \geq 0$$

gives $\cos \delta + \sin \delta \leq (1+h)/\sqrt{1+h^2}$. Therefore a concentric closed side- B core at the chosen net angle has projections strictly smaller than one on both axes of the unit square. It lies strictly inside that square, and hence inside the container. Its mass is at least one by the finite check. Atoms on a core boundary are still strictly inside the parent unit square, so touching unit squares cannot double-count them.

3 Checking every translation exactly

Centre the container at the origin. At a net angle let $c = \cos \theta > 0$ and $s = \sin \theta \geq 0$. The admissible core centres form $[-H, H]^2$, where $H = (L - B(c+s))/2$. The checker requires $H > 0$ at every direction. In rotated coordinates $u = cx + sy$, $v = -sx + cy$, the centre domain is

$$D = \{(u, v) : |cu - sv| \leq H, |su + cv| \leq H\}.$$

An atom contributes its weight on a closed axis-parallel rectangle of side B centred at its rotated coordinates. Rectangle edges and the extreme u coordinates of D partition the plane into open cells of constant mass.

For each open u slab (a, b) intersecting the interior of D , the projection of that intersection on the v axis is an open interval $(v_{\min}, v_{\max})$. When $s > 0$, the lower and upper boundaries of D are

$$\ell(u) = \max \left\{ \frac{cu - H}{s}, \frac{-H - su}{c} \right\}, \quad r(u) = \min \left\{ \frac{cu + H}{s}, \frac{H - su}{c} \right\}.$$

Their minimum and maximum occur at $u = H(c - s)$ and $u = -H(c - s)$, respectively, clamped to $[a, b]$. These evaluations give $v_{\min}$ and $v_{\max}$; for $s = 0$ they are $-H, H$. A v cell is reachable exactly when its open interval overlaps $(v_{\min}, v_{\max})$. A range-minimum sweep checks its mass.

Boundary placements also have mass at least one. Each is a limit of interior placements avoiding all event lines. There are finitely many cells, so a subsequence lies in one fixed cell. Each atom captured there remains captured at the limit because core membership is closed. Nonnegative weights imply that mass cannot decrease at this boundary. This covers wall contacts, coincident events and vertices, completing the translation argument.

3.1 Integer representation and trust boundary

For a reduced half-tangent p/q put

$$C = q^2 - p^2, \quad S = 2pq, \quad R = q^2 + p^2, \quad c = C/R, \quad s = S/R.$$

Choose an integer scale G making centred atom coordinates X_j, Y_j , $\ell_0 = GL/2$ and $b_0 = GB/2$ integral. In units $1/(R^2G)$ the projected atom coordinates and rectangle half-width are

$$U_j = R(CX_j + SY_j), \quad V_j = R(-SX_j + CY_j), \quad J = b_0R^2.$$

With $H_0 = \ell_0R - b_0(C + S)$, the domain inequalities are $|CU - SV| \leq R^2H_0$ and $|SU + CV| \leq R^2H_0$. All geometry and rational comparisons use unbounded Boost integers. Weights use a common denominator; the checker caps the summed integer numerators at 10^{12} to protect signed 64-bit accumulation.

A segment tree and a direct difference-array/prefix-sum engine evaluate the same geometric cells. Their complete logs must agree. This audits two implementations of accumulation, not two independently authored geometric proofs. The JSON input compiler preserves rationals exactly. The numerical optimizer and its sampled constraints are outside the trusted proof core.

4 Dilation and reproducibility

Put $C_0 = \sqrt{1 + h^2}/(B(1 + h)) > 1$. For every rational $0 < r < C_0$, scale the container, cores and atom coordinates by r , leaving weights unchanged. Inverse scaling preserves core coverage and the scaled cores still satisfy the strict angular inequality. The weighted contradiction excludes a packing at side rL . Taking rational r increasing to C_0 proves $s(17) \geq S_*$. The exact rational comparison $(4.613028635886)^2 < S_*^2$ proves the strict decimal statement in Theorem 1. At C_0 the containment inequality becomes equality, which is why the endpoint statement remains weak.

Certificate quantity	Exact value
Atoms	1,620
Directions	2,881
Total mass	$849899249/500000000 = 16.99798498$
Minimum closed-core mass	$1000002103/1000000000 = 1.000002103$
Centre slabs	9,116,871

The authoritative atoms are in `certificates/lower_bound_4p613/best-certificate.json`, with SHA-256

749f13335980a66a27a2304f212b5d8796390c81b962149647a4d9ff43a228ec.

From the repository root, run

```
python3 certificates/lower_bound_4p613/verify.py -source
```

This rebuilds both engines, runs thirteen positive/refusal controls per engine, replays the pinned Levy source and new certificate, compares logs, and checks the result ledger and endpoint. It also runs eight controls for the exact dictionary diagnostic and reconstructs the orbit-deletion experiment in both engines. Python 3.10 or newer with its standard library, C++17 and Boost headers suffice. The command `bash ./verify_all.sh` additionally replays the preceding weighted certificate and regenerates and checks the historical triangle-witness certificate. The weighted package has its own SHA-256 manifest; the historical package retains its original raw and compressed certificate hashes.

5 Spatial search and exact diagnostics

The search alternates between adding under-covered core placements as constraints and using linear-program dual multipliers to propose new point orbits. Final normalization and upward rational rounding produce the measure checked above. These numerical operations suggest data; the full exact replay establishes coverage without assuming numerical feasibility or optimality. An unsuccessful search at side 4.615 supplied some useful support geometry for the final 4.613 run, but supplies no theorem at 4.615.

Two exact diagnostics explain the role of the new sites. For a prescribed 356-orbit dictionary at side 4.61, the package gives 106 contained rational cores C_i and nonnegative rational multipliers z_i . If $A_{ij} = |O_j \cap C_i|$, the diagnostic checks

$$\sum_i z_i A_{ij} \leq |O_j| \quad \text{for every prescribed orbit } O_j, \quad \sum_i z_i = 17.009583872717.$$

Any nonnegative per-atom orbit weights w_j covering these cores must satisfy

$$\sum_j |O_j| w_j \geq \sum_i z_i \sum_j A_{ij} w_j \geq \sum_i z_i > 17.$$

Thus reweighting this dictionary cannot pass the specified core test at 4.61. The conclusion does not constrain other supports or verification methods. The diagnostic recomputes core containment, incidences and dual inequalities using integers and exact fractions; it does not trust a stored incidence matrix.

Separately, the orbit ledger reconstructs the final accepted measure exactly. Deleting its 19 final-stage priced orbits removes 152 atoms of total mass 0.315664656. The remaining mass is 16.682320324, and exact replay rejects at direction zero with core minimum 0.939043209. Uniform rescaling to repair that minimum would exceed mass 17. This concerns deletion from this fixed certificate; reoptimization without those orbits is not ruled out.

Attribution and status. Levy’s original and the derived certificate data are credited under CC BY 4.0; source hashes and changes are recorded in the package’s `ATTRIBUTION.md`. This is a computer-assisted working preprint, with external peer review pending. The earlier computation and exposition were developed with assistance from OpenAI’s GPT-5.6 Pro under human direction; the September patch and this integration also used AI assistance. The claim rests on the explicit data, mathematical reduction and executable checker. No optimality of the measure or the cited upper bound is asserted.

A Historical triangle-witness theorem

The August certificate proves the following earlier result. Throughout this appendix, $L = 4468292/1000000 = 4.468292$.

Theorem 2. *No packing of seventeen unit squares exists in Q_L . Consequently, $s(17) > 4.468292$.*

The preceding exact point-witness bound was $s(17) > 4.456575$, due to Fort[6]. This certificate improves it by 0.011717 and adds orientation-independent triangle witnesses to the subdivision language. The historical data and three checkers remain in `certificates/lower_bound_4p468292/`.

B Strictly unavoidable points

Definition 3. A finite set $P \subseteq Q_L$ is *strictly unavoidable for unit squares* if every unit square $U \subseteq Q_L$ satisfies

$$P \cap \text{int}(U) \neq \emptyset.$$

Lemma 4 (Pigeonhole lemma). *If Q_L contains a strictly unavoidable set of $n - 1$ points, then no packing of n pairwise interior-disjoint unit squares exists in Q_L .*

Proof. Suppose $U_1, \dots, U_n$ were such a packing. For each i , choose $p_i \in P \cap \text{int}(U_i)$. If $p_i = p_j$ for $i \neq j$, then that point lies in both square interiors, contradicting interior-disjointness. Thus the map $i \mapsto p_i$ is an injection from an n -element set into an $(n - 1)$ -element set, which is impossible. $\square$

The use of interiors is important. Boundary witnesses could be shared by several touching squares and would not imply the injection in Lemma 4.

For Theorem 2, the set P consists of the sixteen rational points in Table 1. Every displayed decimal is exact, with denominator 10^7 .

C Parameterizing a unit square

A unit square is unchanged by a rotation through $\pi/2$. We may therefore represent every orientation by an angle

$$-\frac{\pi}{4} \leq \theta \leq \frac{\pi}{4}, \quad t = \tan \theta \in [-1, 1].$$

Let (x, y) be the center of the square, and let $p = (p_x, p_y)$ be a fixed point. Put

$$\Delta x = p_x - x, \quad \Delta y = p_y - y, \quad r = \sqrt{1 + t^2}.$$

In coordinates aligned with the square, the displacement from the center to p is

$$\left(\frac{\Delta x + t\Delta y}{r}, \frac{-t\Delta x + \Delta y}{r} \right).$$

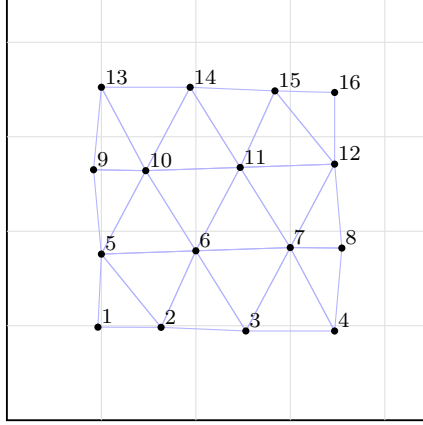
Consequently, p lies strictly inside the square exactly when

$$4(\Delta x + t\Delta y)^2 < 1 + t^2, \tag{2}$$

$$4(-t\Delta x + \Delta y)^2 < 1 + t^2. \tag{3}$$

Table 1: The sixteen rational witness points.

i	x_i	y_i	i	x_i	y_i
1	0.9633740	0.9849850	9	0.9172158	2.6508741
2	1.6317845	0.9829523	10	1.4694893	2.6410332
3	2.5287879	0.9446006	11	2.4689011	2.6753176
4	3.4682923	0.9443212	12	3.4682923	2.7101964
5	0.9999997	1.7580956	13	0.9999997	3.5239708
6	1.9993909	1.7929744	14	1.9395041	3.5236914
7	2.9988027	1.8272588	15	2.8365087	3.4853407
8	3.5451063	1.8219620	16	3.4687946	3.4687946



$$L = 4.468292$$

Figure 1: The witness set and the eighteen certified triangles. The proof uses the exact rational coordinates in Table 1; blue edges indicate the orientation-independent triangle witnesses.

The axis-parallel half-extent of a unit square with parameter t is

$$h(t) = \frac{1 + |t|}{2\sqrt{1 + t^2}}.$$

Thus containment in Q_L requires

$$h(t) \leq x, y \leq L - h(t).$$

The function $u \mapsto (1 + u)/(2\sqrt{1 + u^2})$ is nondecreasing on $[0, 1]$. This monotonicity supplies a simple exact test for pose boxes that cannot contain any feasible square.

D The subdivision certificate

The pose space is parameterized by (x, y, t) . The checker begins with the rational box

$$\mathcal{B}_0 = \left[\frac{1}{2}, L - \frac{1}{2} \right]^2 \times [-1, 1],$$

which contains every feasible pose. It recursively bisects boxes in one of the three coordinates. Each node is encoded by one byte:

Byte	Meaning
0	the current box contains no feasible square pose;
1, ..., 16	the indicated point lies strictly inside every pose in the box;
17, 18, 19	bisect the box in x , y , or t , respectively.
20, ..., 37	the indicated triangle pierces every pose in the box.

The verifier distrusts every leaf claim and recomputes it from the box endpoints.

D.1 Infeasible leaves

For a box

$$\mathcal{B} = [x_0, x_1] \times [y_0, y_1] \times [t_0, t_1],$$

let

$$u = \min\{|t| : t \in [t_0, t_1]\}.$$

Since $h(t) \geq h(u)$ throughout the box, the box is infeasible if any one of

$$x_1 < h(u), \quad L - x_0 < h(u), \quad y_1 < h(u), \quad L - y_0 < h(u)$$

holds. The checker clears denominators and squares nonnegative quantities, so this test is performed using integers only.

D.2 Point-witness leaves

Fix a witness point p . Exact interval arithmetic over $\mathcal{B}$ gives intervals containing

$$A = \Delta x + t\Delta y, \quad B = -t\Delta x + \Delta y.$$

Write M_A and M_B for the maximum absolute endpoint of these intervals. Because $1 + t^2 \geq 1 + u^2$ throughout the box, the two strict inequalities

$$4M_A^2 < 1 + u^2, \quad 4M_B^2 < 1 + u^2$$

prove simultaneously that p is strictly inside every square represented by that box. Again, the implementation stores rational endpoints on fixed grids and clears every denominator before comparison.

The interval enclosure for the bilinear term $t\Delta y$ is obtained by taking the minimum and maximum of its four corner products; the same is done for $(-t)\Delta x$. No floating-point approximation enters a leaf decision.

D.3 Triangle-witness leaves

Some center boxes can be closed without subdividing the orientation interval. The certificate fixes eighteen triangles whose vertices belong to the witness set and applies the following lemma.

Lemma 5 (Strict triangle piercing). *Let A, B, C be three points whose pairwise distances are strictly less than one. Every unit square whose center lies strictly inside triangle ABC contains at least one of A, B, C strictly in its interior.*

Proof. Translate the square center to the origin and rotate the plane so the square is $[-1/2, 1/2]^2$. Its diagonals divide the plane into right, top, left, and bottom closed cones. Because the origin lies strictly inside triangle ABC , the three vertices cannot all lie in a closed half-plane bounded by either diagonal. Among the four cones this forces two vertices into opposite cones. If neither vertex were in the square interior, vertices in the right and left cones would have x -coordinates differing by at least one; vertices in the top and bottom cones would have y -coordinates differing by at least one. In either case their Euclidean distance would be at least one, contradicting the strict side-length hypothesis. $\square$

For a triangle leaf, the checker first verifies the three strict inequalities

$$\|A - B\|_2^2 < 1, \quad \|B - C\|_2^2 < 1, \quad \|C - A\|_2^2 < 1.$$

It orients the triangle counterclockwise and checks that all four corners of the center rectangle lie strictly to the left of each directed edge. Each edge test is affine in the center coordinates, so positivity at the four corners proves positivity throughout the rectangle. Lemma 5 then covers every orientation in the box. The checker clears denominators and performs these tests with integers only.

D.4 Completeness of the tree

The checker traverses the tree from $\mathcal{B}_0$, applying every split exactly. It rejects an early end of file, an unknown opcode, a false leaf claim, or any trailing byte. Finally it checks the full-binary-tree identity

$$\#\text{leaves} = \#\text{split nodes} + 1.$$

It follows inductively that the accepted leaves form a cover of the complete root box. Every feasible pose is therefore covered by either a point-witness leaf or a triangle-witness leaf. In both cases at least one of the sixteen points lies strictly inside the square, so the point set is strictly unavoidable.

Lemma 4 now rules out a packing in Q_L , and therefore in every smaller container as well. Thus $s(17) \geq L$. The minimum defining $s(17)$ is attained by compactness, so equality would give a packing in Q_L . Hence $s(17) > L$, proving Theorem 2.

E Exact arithmetic and reproducibility

The certificate uses the scales

$$S_t = 2^{40}, \quad S_x = 10^6 2^{40}, \quad D_p = 10^7.$$

The center and slope endpoints are stored as integers on these fixed rational grids, while the witness points have denominator D_p . The accepted tree has

Total nodes	122,626,747
Split nodes	61,313,373
Point-witness leaves	33,762,069
Triangle-witness leaves	483,875
Infeasible leaves	27,067,430
Maximum depth	74

The certificate consists of one byte per node. Its raw SHA-256 digest is

2838f315302d67da131745925e9ec7dd2a602bb299d1335ce25e4e13a7b7b6d2.

The raw tree is 122,626,747 bytes, above GitHub’s ordinary per-file limit. The repository stores a deterministic 374,096-byte XZ archive with SHA-256

`a349b81e630ccf7292ae0afe6ed954591f7e88fe6289847f67883204a7ed60ac.`

The repository provides three checkers:

1. a fast C++ checker using bounded wide integers and the same arithmetic header as the generator;
2. a separately implemented C++ checker using arbitrary-precision integers;
3. a separately implemented Python checker using arbitrary-precision integers.

The generator and its search heuristics are not part of the trusted proof core: each checker reads the certificate as untrusted input and recomputes every leaf claim. The two arbitrary-precision checkers do not reuse the generator’s arithmetic routines, and so do not rely on a fixed-width overflow analysis. From the repository root, the command

`bash ./verify_all.sh`

replays the weighted certificate and regenerates the historical certificate, checks its hash, compares it byte-for-byte with the archive, and executes the three historical verifiers. The proof package is available at <https://github.com/Mira-acc/17squares>.

References

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